

• Mean: Let X be a random variable. The mean of X , denoted by μ_X or $E(X)$, is defined by

(i) $E(X) = \sum x_i p_X(x_i)$ if X is discrete with mass points $x_1, x_2, \dots, x_j, \dots$

(ii) $E(X) = \int_{-\infty}^{\infty} x f_X(x) dx$ if X is continuous with probability density function $f_X(x)$.

(iii) $E(X) = \int_{-\infty}^{\infty} [1 - F_X(x)] dx = \int_{-\infty}^0 F_X(x) dx$ for an arbitrary random variable X .

Remark: 1. $E(X)$ is the center of gravity (or centroid) of the unit mass that is determined by the density function of X . So the mean of X is a measure of where the values of the random variable X are "centered".

2. The mean $E(X)$ exists provided the series (i) or the integral (ii) are absolutely convergent.

Example 1: Let X be a continuous RV with pdf $f_X(x) = \lambda e^{-\lambda x}$, $x > 0$, $\lambda > 0$

$$\therefore E(X) = \int_{-\infty}^{\infty} x f_X(x) dx = \int_0^{\infty} x \lambda e^{-\lambda x} dx = \left(\frac{1}{\lambda} \right) \checkmark$$

Example 2: Let X be a random variable with pdf $f_X(x) = \frac{1}{x^2}$, $x > 1$; then

$$E(X) = \int_1^{\infty} x \cdot \frac{dx}{x^2} = \lim_{b \rightarrow \infty} \int_1^b \frac{dx}{x} = \lim_{b \rightarrow \infty} \log_e b = \infty,$$

$E(X)$

so, we say that $E(X)$ does not exist.

$$Y = \frac{1}{2} m X^2$$

$E(Y)$

$$E\left(\frac{1}{2} m X^2\right)$$

• Mean/ Expected Value of a Function of a Random Variable:

Let X be a random variable and g be a function with both domain and range the real line. The expectation or expected value of the function $g(\cdot)$ of the RV X , denoted by $E[g(X)]$, is defined by

(i) $E[g(X)] = \sum_i g(x_i) p_X(x_i)$ if X is discrete with mass points $x_1, x_2, \dots, x_j, \dots$ (provided this series is absolutely convergent.)

(ii) $E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$ if X is continuous with pdf $f_X(x)$ (provided the integral is absolutely convergent).

Theorem: Below are properties of expected value

(i) $E[c] = c$ for a constant c

$$E(c) = \int_{-\infty}^{\infty} c f_X(x) dx = c \int_{-\infty}^{\infty} f_X(x) dx = c \checkmark$$

(ii) $E[cg(X)] = cE[g(X)]$ for a constant c

$\text{Var}(X)$

(iii) $E[c_1 g_1(X) + c_2 g_2(X)] = c_1 E[g_1(X)] + c_2 E[g_2(X)]$

$$= E(X - \mu)^2$$

$$= E(X^2 - 2\mu X + \mu^2)$$

$$= E(X^2) - 2\mu E(X) + \mu^2$$

(iv) $E[g_1(X)] \leq E[g_2(X)]$ if $g_1(x) \leq g_2(x)$ for all x

• Variance: Let X be a random variable, and $\mu_X = E(X)$. The variance of X , denoted by σ_X^2 or $\text{var}(X)$, is defined by

$$\text{Var}(X) = E(X - \mu_X)^2$$

$$= \begin{cases} \sum_i (x_i - \mu_X)^2 p_X(x_i) & \text{if } X \text{ is discrete with pmf } p_X(x) \\ \int_{-\infty}^{\infty} (x - \mu_X)^2 f_X(x) dx & \text{if } X \text{ is continuous RV with pdf } f_X(x). \end{cases}$$

$$= E(X^2) - 2\mu\mu + \mu^2 = E(X^2) - \mu^2$$

$$\text{Var}(X) = E(X^2) - \{E(X)\}^2$$

Theorem: If X is a random variable, $\text{var}(X) = E[X - E(X)]^2 = E[X^2] - \{E(X)\}^2$.

Proof: $E[X - E(X)]^2 = E[X^2 - 2XE(X) + \{E(X)\}^2]$

$$= E[X^2] - 2[E(X)]^2 + (E(X))^2$$

$$\therefore \text{Var}(X) = E(X^2) - \{E(X)\}^2 \checkmark$$

Example: Let X be a random variable with pdf $f(x) = \begin{cases} \frac{1}{\beta-\alpha} & \text{if } \alpha \leq x \leq \beta \\ 0 & \text{otherwise} \end{cases}$

Find the mean and variance of X .

Solⁿ: The mean of the random variable X is, $E(X) = \int_{-\infty}^{\infty} x f_X(x) dx = \int_{\alpha}^{\beta} \frac{x}{\beta-\alpha} dx = \frac{\alpha+\beta}{2}$.

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f_X(x) dx = \int_{\alpha}^{\beta} \frac{x^2}{\beta-\alpha} dx = \frac{\beta^3 - \alpha^3}{3(\beta-\alpha)} = \frac{\beta^2 + \alpha\beta + \alpha^2}{3}$$

Hence, $\text{Var}(X) = E(X^2) - \{E(X)\}^2$

$$= \frac{\beta^2 + \alpha\beta + \alpha^2}{3} - \left(\frac{\alpha+\beta}{2}\right)^2$$

$$= \frac{(\beta-\alpha)^2}{12}$$

Example: Let X be a random variable with pdf $f_X(x) = \lambda e^{-\lambda x}$, $x > 0$ and 0 otherwise. Find $\text{var}(X)$.

Solⁿ: $E(X) = \int_0^{\infty} x \lambda e^{-\lambda x} dx = \frac{1}{\lambda}$

$$E(X^2) = \int_0^{\infty} x^2 \lambda e^{-\lambda x} dx = \frac{2}{\lambda^2}$$

$$\therefore \text{var}(X) = E(X^2) - (E(X))^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$$

• Standard deviation: If X is a random variable, the standard deviation of X , denoted by σ_X , is defined as $\sqrt{\text{var}(X)}$.

• Moments and Moment Generating Functions

If X is a random variable, the n th moment of X , usually denoted by μ'_n , is defined as $\mu'_n = E[X^n]$ if the expectation exists.

Note that $\mu'_1 = E(X) = \mu_X$, the mean of X .

Central moments: n th order central moment $\mu_n = E[X - \mu_X]^n$

Note that $\mu_1 = E[X - \mu_X] = E(X) - \mu_X = 0$ and $\mu_2 = E(X - \mu_X)^2$, the variance of X .

• Moment generating function^(mgf): Let X be a random variable with pmf $p_X(x)$ or pdf $f_X(x)$ depending on discrete and continuous. The expected value of e^{tx} is defined to be the moment generating function of X if the expected value exists for every value of t in some interval $-h < t < h$; $h > 0$. The mgf, denoted by $M_X(t)$, is

$$M_X(t) = E(e^{tx})$$

$$= \begin{cases} \int_{-\infty}^{\infty} e^{tx} f_X(x) dx & \text{if the RV } X \text{ is continuous} \\ \sum_x e^{tx} p_X(x) & \text{if the RV } X \text{ is discrete} \end{cases}$$

Remark: We shall show that

$$\mu'_n = E(X^n) = \left. \frac{d^n}{dt^n} M_X(t) \right|_{t=0}$$

Example: Let X be a random variable with pdf $f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$

Find the MGF of X and hence find the mean and variance of X .

Solⁿ: $M_X(t) = E(e^{tx}) = \int_0^{\infty} e^{tx} \cdot \lambda e^{-\lambda x} dx = \frac{\lambda}{\lambda - t}$ for $t < \lambda$

$$M'_X(t) = \frac{\lambda}{(\lambda - t)^2} \Rightarrow E(X) = M'_X(0) = \frac{1}{\lambda}$$

$$M''_X(t) = \frac{2\lambda}{(\lambda - t)^3} \Rightarrow E(X^2) = M''_X(0) = \frac{2}{\lambda^2} \therefore \text{var}(X) = E(X^2) - \{E(X)\}^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$$